

Enhanced formic acid production for CO₂ photocatalytic reduction over Pd/H-TiO₂ catalyst

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Abstract The photocatalytic reduction of CO₂ into formic acid is a feasible approach to alleviate the effects of global climate change and achieve chemical energy storage. It is important to design highly active photocatalysts to improve the selectivity and yield of formic acid. In this study, TiO₂-based catalysts were prepared and loaded with Pd nanoparticles via an impregnation process. The Pd/H-TiO₂ catalyst demonstrated superior CO₂ reduction activity and a high formic acid production rate of 14.14 mmol_{cat}·g⁻¹·h⁻¹. The excellent catalytic performance observed in the presence of a Pd/H-TiO₂ catalyst is ascribed to the synergy between O_v and Pd. The presence of O_v led to increase in CO₂ adsorption while Pd loading enhanced the photogenerated electron-hole pair separation. Electron transfer from H-TiO₂ to Pd also contributed to CO₂ activation.

Keywords CO₂ reduction, formic acid, photocatalysis, TiO₂ catalyst

1 Introduction

CO₂ emissions from fossil fuel combustions result in numerous severe environmental problems [1]. The conversion of CO₂ into carbon-containing solar fuels is a promising technology to alleviate the harmful effects of global climate change. CO₂ can be converted into various chemicals, including CO, CH₃OH, HCOOH, and CH₄ [2]. Among them, HCOOH is widely studied owing to its considerable market demand and high energy density [3]. For example, it possesses a remarkable volumetric hydrogen storage density of 53 g·L⁻¹, which is equivalent to 600 times of compression hydrogen; as such, HCOOH

can be used as a hydrogen storage molecule [4]. Several pathways exist for the conversion of CO₂ into chemicals [5], including photocatalytic reduction, thermochemical reduction, biochemical reduction, and electrochemical reduction. The photocatalytic reduction of CO₂ does not require further thermal energy input; it conducts a conversion reaction through the transfer of electrons generated by the catalyst being excited by light to CO₂. However, the photocatalytic reduction of CO₂ is a multiple-electron transfer process involving complex reaction pathways and yields different products depending on the number of electrons transferred [6], resulting in poor product selectivity and a low efficiency in producing HCOOH. Therefore, to achieve highly selective HCOOH synthesis and increase the production rate of HCOOH, it is imperative to design photocatalysts with conduction potentials close to the redox potential of HCOOH.

Many inorganic semiconductor materials, such as doped Al₂O₃ [7], ZnO, and TiO₂ have been applied to CO₂ photocatalytic reduction for producing HCOOH [8]. Among them, TiO₂ is one of the most popular materials and exhibits excellent photocatalytic performance owing to its suitable redox potential. However, TiO₂ exhibits a low photocatalytic reduction efficiency owing to its large band gap and rapid electron-hole recombination [9]. Furthermore, TiO₂ has a high redox potential, resulting in the occurrence of side reactions and difficulties in obtaining HCOOH with high selectivity. To overcome these limitations, several strategies have been studied, including metal loading, semiconductor coupling, dye sensitization, and construction defect engineering [10]. Among them, metal loading and vacancy defect construction in semiconductors are two effective approaches to slow down electron-hole recombination [11].

Research has shown that the formation of oxygen vacancies (O_v) considerably affects photocatalytic CO₂ reduction performance [12,13]. The formation of O_v increases CO₂ adsorption and thereby enhances

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photocatalytic CO₂ reduction efficiency. Zhao et al. [14] summarized how defective sites improve the photocatalytic reduction of CO₂ and confirmed that formation of O_v on the TiO₂ surface can enhance the adsorption and conversion of CO₂. However, merely promoting CO₂ adsorption is insufficient to enhance the photocatalytic CO₂ reduction efficiency. Efficient photoelectron transfer is another key factor in improving photocatalytic CO₂ reduction performance. The rapid photogenerated electron-hole recombination in TiO₂ causes low photocatalytic CO₂ reduction activity [15]. Modifying TiO₂ with metal is not only an effective and direct method to reduce the band gap of TiO₂ but can also slow down the recombination of photogenerated electron-hole pairs, thereby resulting in improved photocatalytic activity. Recently, metal cocatalysts, such as Pt [16], Pd [17], and Ag [18] have been employed in photocatalytic CO₂ reduction. The Schottky junction formed by loading metals on semiconductors can trap photogenerated electrons, after which the separated electrons are transferred to the surface of the catalyst for CO₂ conversion, thereby considerably increasing the photocatalytic efficiency [19]. Li et al. [11] designed a Ag/TiO₂ catalyst via a hydrothermal process involving *in situ* photodeposition; their results demonstrated increased photocatalytic CO₂ reduction activity, which was attributed to the combined effect of Ag and defective TiO₂.

Herein, we prepared TiO₂-based catalysts and investigated how O_v and Pd influenced CO₂ adsorption and activation and thereby improved the photocatalytic activity. Hydrogen pretreatment was found to lead to the incorporation of H atoms into O_v, forming Ti–H bonds to guarantee the high selectivity of HCOOH. Characterizations and density functional theory (DFT) calculations confirmed that O_v increases the CO₂ adsorption capacity of TiO₂ and that loaded Pd particles can trap photogenerated electrons from semiconductors to promote electron-hole separation.

2 Experimental

2.1 Materials

Anatase TiO₂ (99.8%), PdCl₂ (99.9%), NaBH₄ (98%), K₂CO₃ (98%), KOH (95%), KHCO₃ (99.5%), and a standard HCOOH solution (analytic grade) were purchased from Shanghai Aladdin Reagent Co., China.

2.2 Catalyst synthesis

Anatase TiO₂ was employed as a precursor without any treatment (hereafter referred to as TiO₂). Hydrogenated TiO₂ (H-TiO₂) was obtained by pretreating TiO₂ in a 5 vol % H₂/Ar atmosphere at 500 °C for 12 h. Based on the pretreatment temperature, the obtained catalysts were

denoted as H-TiO₂-T (where T represents hydrogenation temperature).

The Pd/H-TiO₂ catalyst was fabricated via impregnation [20]. In particular, PdCl₂ (0.0284 mmol), H-TiO₂ (0.5 g), and deionized water (20 mL) were mixed in a beaker. The obtained substance was stirred for 24 h at 25 °C until a well-dispersed suspension was obtained; subsequently, a 1 mol·L⁻¹ aqueous NaOH solution was used to adjust the pH of the suspension to 10.0–11.0. A 30 mg·mL⁻¹ aqueous NaBH₄ solution was added to the obtained alkaline suspension at a molar ratio of $n(\text{Pd}):n(\text{NaBH}_4) = 1:6$, followed by stirring for 2 h until no bubble production occurred. A solid substance was separated from the resultant suspension via filtration and washed with water to remove any residual material; subsequently, the washed solid substance was baked at 110 °C for 12 h to produce a sample, which was termed Pd/H-TiO₂.

2.3 Catalyst characterization

The crystalline structures of the TiO₂, H-TiO₂, and Pd/H-TiO₂ catalysts were analyzed via X-ray diffraction (XRD, Rigaku Ultima IV). The Pd content of the Pd/H-TiO₂ catalyst was monitored via inductively coupled plasma-optical emission spectrometry (ICP-OES; Agilent 5110). Transmission electron microscopy (TEM) and high-resolution TEM (HRTEM) were performed using a JEOL JEM 2100F transmission electron microscope to determine the morphology and metal particle size of the synthesized catalysts. The electronic valence states of the elements in the prepared catalysts were evaluated via X-ray photoelectron spectroscopy (XPS, Thermo Scientific K-Alpha). The photoelectric chemical properties of the prepared catalysts were analyzed using an electrochemical workstation (CHI 760E). Photoluminescence spectra (PL spectra) were obtained to analyze the separation efficiency of the photogenerated electron-hole pairs. FLUKE infrared imaging was used to detect the temperature distribution of the sample surface in a noncontact manner throughout this process, and images were obtained. O_v characterization of the catalyst surface was performed via electron paramagnetic resonance spectrometry (EPR, Bruker EMX PLUS). The CO₂ adsorption capacity of the catalyst was recorded via CO₂ temperature-programmed desorption (CO₂-TPD, Micromeritics AutoChem II 2920).

2.4 CO₂ hydrogenation performance

The photocatalytic reduction of CO₂ was performed at 30 °C and 2.0 MPa in an autoclave reactor (50 mL, Anhui Kemi Machinery Technology Co., Ltd.) equipped with a sapphire window. Deionized water (10 mL), KOH (10 mL), and catalyst (2 g·L⁻¹) were mixed in the autoclave reactor, which was then sealed. The residual air

was displaced with CO₂ three times. Subsequently, 1.0 MPa CO₂ and 1.0 MPa H₂ were fed into the sealed reactor and 300 nm ultraviolet (UV) light was used as the light source for vertical exposure through the window inside the reactor. Temperature was precisely controlled using the thermocouple of the tank reactor. The experimental diagram is shown in Fig. S1 (cf. Electronic Supplementary Material, ESM). The resulting liquid products were detected using high-performance liquid chromatography (Agilent 1260). The rate of HCOOH production was calculated using Eq. (1) as follows:

$$Y = \frac{n_{\text{HCOOH}}}{m_{\text{cat.}} \times t}, \quad (1)$$

where n_{HCOOH} represents the amount of HCOOH after the reaction (mmol), $m_{\text{cat.}}$ represents the catalyst mass (g), and t represents the reaction time (h).

2.5 Theoretical calculations

Simulation calculations based on DFT were used to investigate the effect of O_v and metal Pd loading on CO₂ adsorption and activation. The TiO₂ (101) surface was modeled based on HRTEM characterization. All calculations were performed using the Vienna Ab initio Simulation Package, applying the projector augmented wave (PAW) method [21,22]. Interactions between core and valence electrons was analyzed by PAW. To determine the total energy, a cutoff energy of 400 eV was set for plane wave basis sets, through which the valence electronic states were expanded. The convergence criterion for the self-consistent electronic loop was set at 10⁻⁵ eV. When the maximum force received by the atom was less than 0.02 eV·Å⁻¹, the calculation was considered to have converged. The Monkhorst-Pack k network was specified as 2 × 2 × 1 on the TiO₂ (101) surface with a vacuum thickness of 15 Å. The adsorption energy (E_{ads}) of CO₂ on the catalyst was calculated using Eq. (2):

$$E_{\text{ads}} = E_{\text{total}} - E_{\text{catalyst}} - E_{\text{CO}_2}, \quad (2)$$

where E_{total} is the total energy of CO₂ adsorbed to Pd metal particles on the support, E_{catalyst} is the total energy of the Pd/H-TiO₂ system, and E_{CO_2} is the total energy of the gas molecules. The charge density difference ($\Delta\rho$) was calculated using Eq. (3):

$$\Delta\rho_{\text{AB}} = \rho_{\text{A}} - \rho_{\text{B}}, \quad (3)$$

where $\Delta\rho_{\text{AB}}$, ρ_{A} , and ρ_{B} denote charge densities of the systems AB (CO₂ adsorbed on Pd/H-TiO₂), A (CO₂ or CO₂-Pd of CO₂ adsorbed on Pd/H-TiO₂), and B (Pd/H-TiO₂ or H-TiO₂ of CO₂ adsorbed on Pd/H-TiO₂), respectively. It is important to note that since the roles played by Pd and TiO₂ in the system are not yet clear, their interactions and the charge transfers between them were considered separately.

3 Results and discussion

3.1 Catalyst properties

Figure 1(a) shows XRD information of TiO₂, H-TiO₂, and Pd/H-TiO₂ catalysts. Changes in the crystalline structure of TiO₂ resulted from pretreatment at different temperatures. When compared with TiO₂, the diffraction peak crystallinity corresponding to the (101) plane decreased as the hydrogenation temperature increased from 400 to 600 °C. The characteristic peak at 27.58° matched with that of rutile TiO₂, indicating that the crystalline structure of H-TiO₂ gradually transformed from anatase to rutile with increasing temperature [23]. Previous investigations revealed that the rutile phase could inhibit the photocatalytic activity [24]. Therefore, Pd/H-TiO₂ was prepared via hydrogen pretreatment at 500 °C. The Pd/H-TiO₂ catalyst exhibited three major characteristic peaks at 25.3°, 37.9°, and 48.1°, which correspond to the (101), (004), and (200) planes, respectively (JCPDS No. 21-1272). The diffraction peaks of Pd were not observed, implying that Pd was highly dispersed on H-TiO₂ [25]. Furthermore, it was clear that the Pd/H-TiO₂ catalyst had characteristic diffraction peaks similar to those of H-TiO₂, indicating that Pd loading did not alter the crystal structure of anatase TiO₂ [26]. The main characteristic peaks, i.e., (101), (004), and (200), exhibited minimal shift, indicating that Pd did not enter the lattice of H-TiO₂; instead, it was highly dispersed on the surface [20]. The micromorphology and crystalline phase of the TiO₂, H-TiO₂, and Pd/H-TiO₂ catalysts were verified using TEM. As shown in Fig. 1(b), the O_v from hydrogen treatment and Pd loading did not affect the morphology, and Pd with an average diameter of 5 nm remained on H-TiO₂ and the TiO₂ catalysts. In addition, the Pd/H-TiO₂ catalyst exhibited a 0.22 nm lattice fringe spacing, as shown in HRTEM images (Fig. 1(c)), corresponding to Pd (111), and the lattice stripe spacing of anatase (101) was calculated to be 0.34 nm. As shown in Fig. 1(d), energy dispersive spectroscopy (EDS) analysis of the Pd/H-TiO₂ catalyst revealed that it contained Ti, O, and Pd, which were evenly distributed across the surface of H-TiO₂. According to the ICP-OES data shown in Table S1 (cf. ESM), the actual amount of Pd in the catalyst was consistent with the theoretical content obtained from the calculations, demonstrating that Pd nanoparticles were successfully loaded on H-TiO₂ using the impregnation method.

To further study the chemical compositions of various elements, including Pd, Ti, and O in the catalysts, the catalysts were examined using XPS. The resulting survey spectrum is shown in Fig. 2(a). Two characteristic peaks were observed at binding energies of 458.7 and 464.0 eV, which were assigned to Ti 2p_{3/2} and Ti 2p_{1/2} in TiO₂,

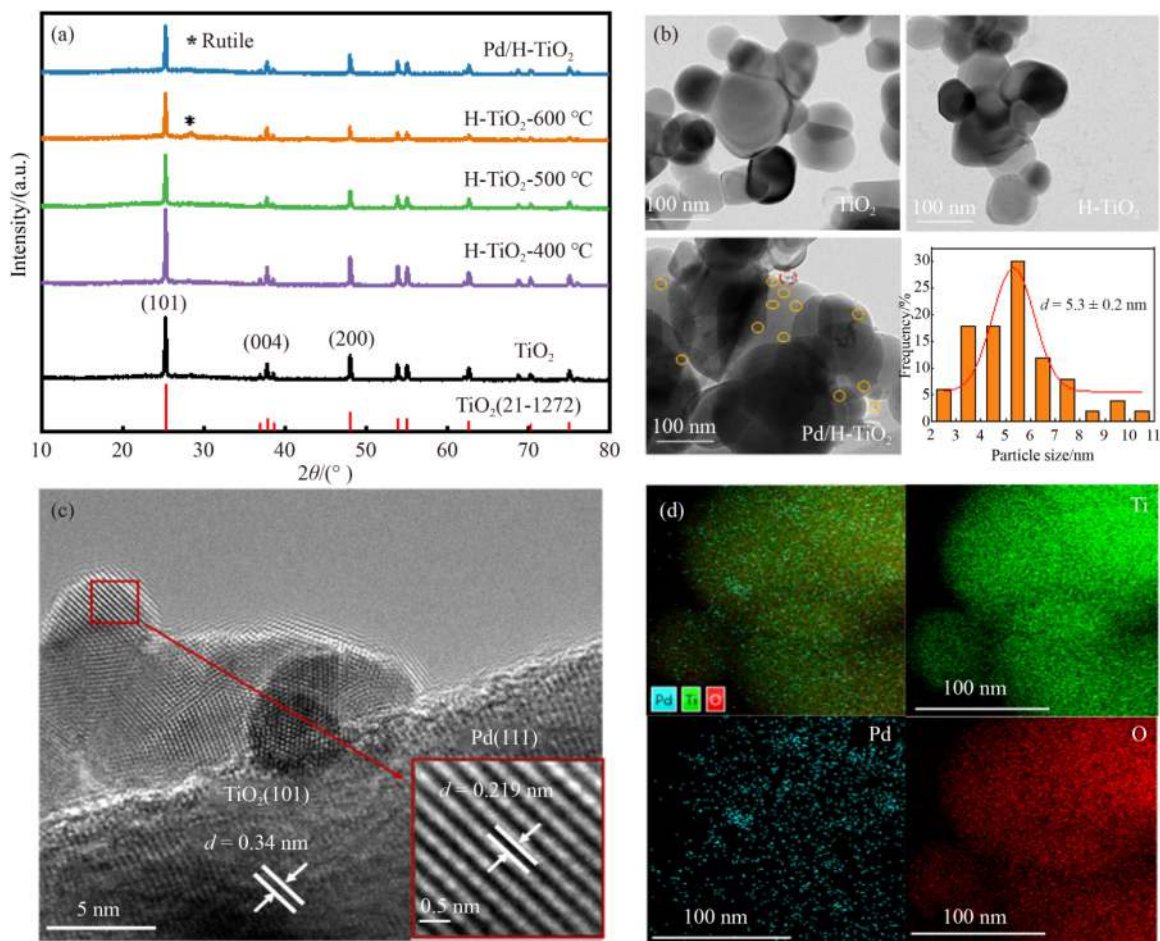


Fig. 1 (a) XRD patterns of TiO₂ at different hydrogenation temperatures and Pd/H-TiO₂, (b) TEM image of catalysts, (c) HRTEM image of Pd/H-TiO₂, (d) EDS elemental mapping of Pd/H-TiO₂.

respectively. In the XPS spectra of O 1s (Fig. 2(b)), the peak centered at ~ 531.7 eV was attributed to O_v . Because the lack of oxygen atoms in the sample can cause a change of chemical state, the metal ions must lower their valence state to remain electrically neutral. Accordingly, upon hydrogen treatment of TiO₂, in addition to Ti^{4+} , Ti^{3+} is formed in H-TiO₂ [27]. This indicates that hydrogen treatment of TiO₂ can produce partial O_v . Furthermore, the binding energy of Ti 2p_{3/2} in H-TiO₂ (458.5 eV) shifts toward lower binding energy relative to that of TiO₂ (458.7 eV), implying the formation of Ti-H bonds in H-TiO₂, which is in accordance with the phenomenon reported by Samsudin et al. [28]. We further conclude that the formation of Ti-H bonds occurs owing to the occupation of O_v by H atoms according to the results reported by Liu et al. [29]. The presence of H atoms can increase the electron density of the adjacent Ti, causing the binding energy to shift to lower energy regions [29,30]. As shown in Fig. 2(c), the peaks located at 334.6 and 339.9 eV match well with those of Pd 3d_{5/2} and Pd 3d_{3/2}, respectively. Additionally, the spectral peaks of Pd were found to have shifted toward lower binding energies relative to the standard binding energy of Pd⁰ (335.4 and 340.7 eV). Moreover, the binding

energy of Ti shifted toward higher energies after Pd loading, signifying an interaction between Pd and H-TiO₂ and the transfer of electrons from H-TiO₂ to Pd [31].

EPR experiments were conducted to detect O_v in various photocatalysts, as shown in Fig. 2(d). The characteristic signal at a g value of 1.973 in TiO₂ denotes the presence of a metal species [32]. Following hydrogen treatment, a distinct characteristic signal was found at the g value of 2.002 for H-TiO₂ and Pd/H-TiO₂, corresponding to O_v on the surface of the material [33]. This result is consistent with the information obtained from XPS analysis.

3.2 Photocatalytic reduction of CO₂

3.2.1 Performance of photocatalytic reduction of CO₂ to HCOOH

The photocatalytic activities of the catalysts were evaluated at 30 °C, 2.0 MPa, and a light intensity of $4.57 \text{ mW} \cdot \text{cm}^{-2}$. It was found that the HCOOH production rate over untreated TiO₂ was only $0.83 \text{ mmol} \cdot \text{g}_{\text{cat}}^{-1} \cdot \text{h}^{-1}$ in Fig. 3(a). Following treatment with H₂, the HCOOH production rate on the H-TiO₂ catalyst reached

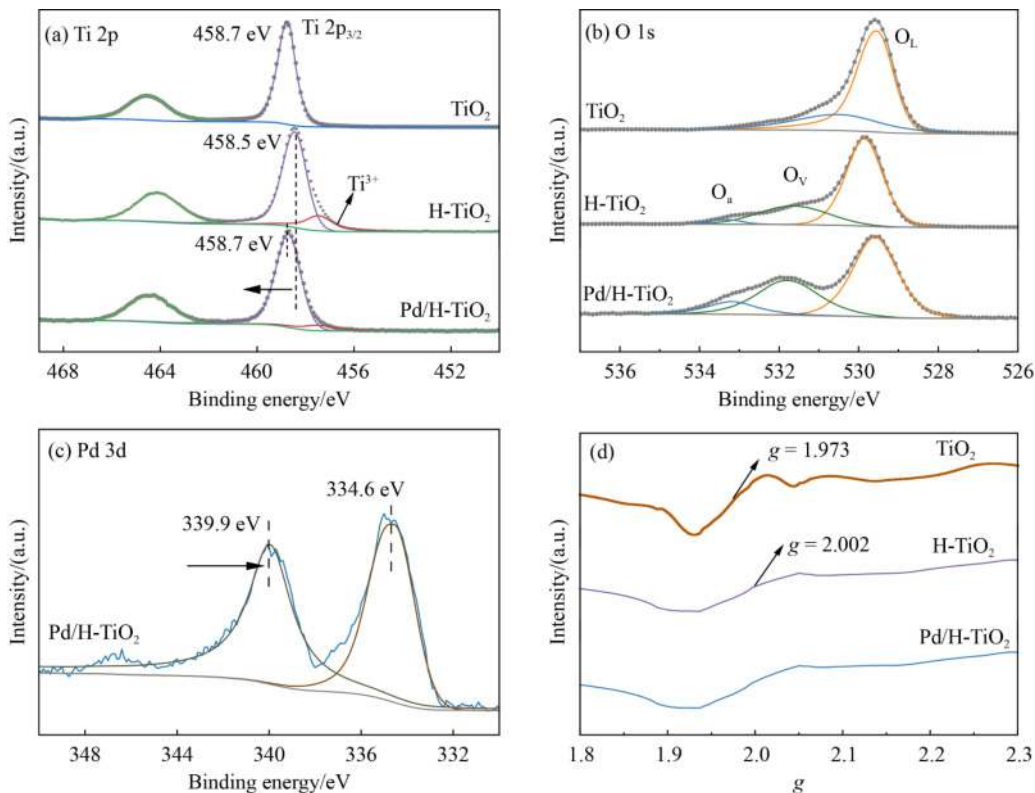


Fig. 2 XPS spectra of (a) Ti 2p, (b) O 1s, (c) Pd 3d, and (d) EPR of TiO₂, H-TiO₂, and Pd/H-TiO₂ obtained at room temperature.

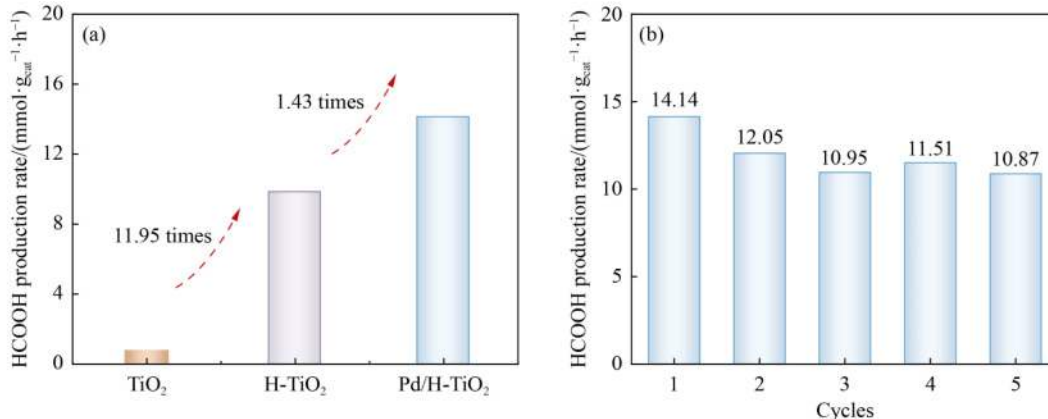


Fig. 3 (a) HCOOH production rate within 1 h with different catalysts, (b) cyclic stability of the Pd/H-TiO₂ catalyst.

9.86 mmol·g_{cat}⁻¹·h⁻¹, i.e., 11.95 times higher than that of TiO₂. Additionally, the reduction activity and HCOOH production rate (14.14 mmol·g_{cat}⁻¹·h⁻¹) of Pd/H-TiO₂ was superior to that of the other catalysts. Compared with TiO₂ and H-TiO₂, the total HCOOH production of Pd/H-TiO₂ also increased significantly within 3 h (Fig. S2, cf. ESM). Throughout the process, the gases before and after the reaction were collected and tested, showing that almost no other carbon-containing products were produced. As such, the selectivity of HCOOH was close to 100%. When the reaction was catalyzed by Pd/H-TiO₂, the HCOOH production rate showed a significant increase, comparable to recent reports (Table S2, cf. ESM) [34,35].

We further analyzed the reasons for the high selectivity and activity of HCOOH when using the Pd/H-TiO₂ catalyst. From XPS results (i.e., the binding energy of Ti in the TiO₂ catalyst was 458.7 eV), the binding energy of Ti 2p_{3/2} was found to shift toward lower values (458.5 eV) after hydrogen pretreatment, suggesting that most O_v was replaced by hydrogen and the Ti-H bond formed for higher HCOOH selectivity. This phenomenon had previously been demonstrated in a semiconductor catalysts system [29,36]. Furthermore, the CO₂ adsorption capacity of the catalyst was analyzed using CO₂-TPD as shown in Table S1. Compared to the TiO₂ catalyst, the adsorption of CO₂ over H-TiO₂ and Pd/H-TiO₂ reached 0.13 and 0.18 mmol·g⁻¹, respectively, demonstrating that

O_v could increase the CO_2 adsorption and further enhance the HCOOH yield.

Cycling experiments were conducted under the same conditions to evaluate the long-term stability of the Pd/H-TiO₂ catalyst. The results of the experiments indicated that the photocatalytic reduction of CO_2 activity decreased slightly after five cycles under UV light irradiation (Fig. 3(b)). To reveal the reason for the decrease of HCOOH yield, TEM analysis (Fig. S3, cf. ESM) was performed for the catalyst after running five cycles. Agglomeration of metal Pd particles was observed; this might cause the reformation of electron-hole recombination sites, leading to a decrease in the effective electrons number involved in the CO_2 conversion reaction. Consequently, a slight decrease in HCOOH production rate was observed after five cyclic reactions.

The optical absorption properties of the Pd/H-TiO₂ photocatalyst were determined using UV-vis diffuse-reflectance spectroscopy as shown in Fig. 4(a). According to the resulting Tauc plots, the band gap (E_g) for Pd/H-TiO₂ was estimated to be 2.7 eV (Fig. 4(b)). Mott-Schottky measurements were determined to investigate the band structure of the Pd/H-TiO₂ catalyst (Figs. 4(c) and 4(d)). The positive slope of the M-S curve confirmed that the prepared Pd/H-TiO₂ catalyst was an *n*-type semiconductor. The conduction band (CB) values of *n*-type semiconductors can be estimated using flat band values. Based on the data in Fig. 4(c), the flat band value of the Pd/H-TiO₂ catalyst with respect to the Ag/AgCl

electrode was -0.96 V. The valence band (VB) values of the Pd/H-TiO₂ catalyst with respect to the normal hydrogen electrode (NHE) were obtained using formulas (4) and (5) [37] as follows:

$$E[\text{vs. NHE}] = 0.2 \text{ V} + E[\text{vs. Ag/AgCl}], \quad (4)$$

$$E_{\text{VB}} = E_g + E_{\text{CB}}. \quad (5)$$

It has been reported that the flat potential of *n*-type semiconductors was more positive by 0.1 V relative to the CB [38]. The flat band for the Pd/H-TiO₂ catalyst was -0.76 V vs. NHE. Thus, the calculated E_{CB} was -0.86 V vs. NHE and the E_{VB} of the Pd/H-TiO₂ catalyst was 1.84 V. This result demonstrates that Pd/H-TiO₂ catalyst meets the redox potential requirements for the photocatalytic reduction of CO_2 to HCOOH.

3.2.2 Microscopic interactions

To analyze the influence of O_v and Pd on CO_2 adsorption, the adsorption energy and differential charge density of CO_2 on TiO₂, H-TiO₂, and Pd/H-TiO₂ surfaces were studied using DFT calculations. Based on various experimental characterization methods, TiO₂, H-TiO₂, and Pd/H-TiO₂ models were constructed in Fig. S4 (cf. ESM) to reveal the mechanisms of interaction in CO_2 reduction systems. Due to differences in the coordination environments of Ti, oxygen vacancy sites were considered in terms of O_{v1} and O_{v2} . The structural information and

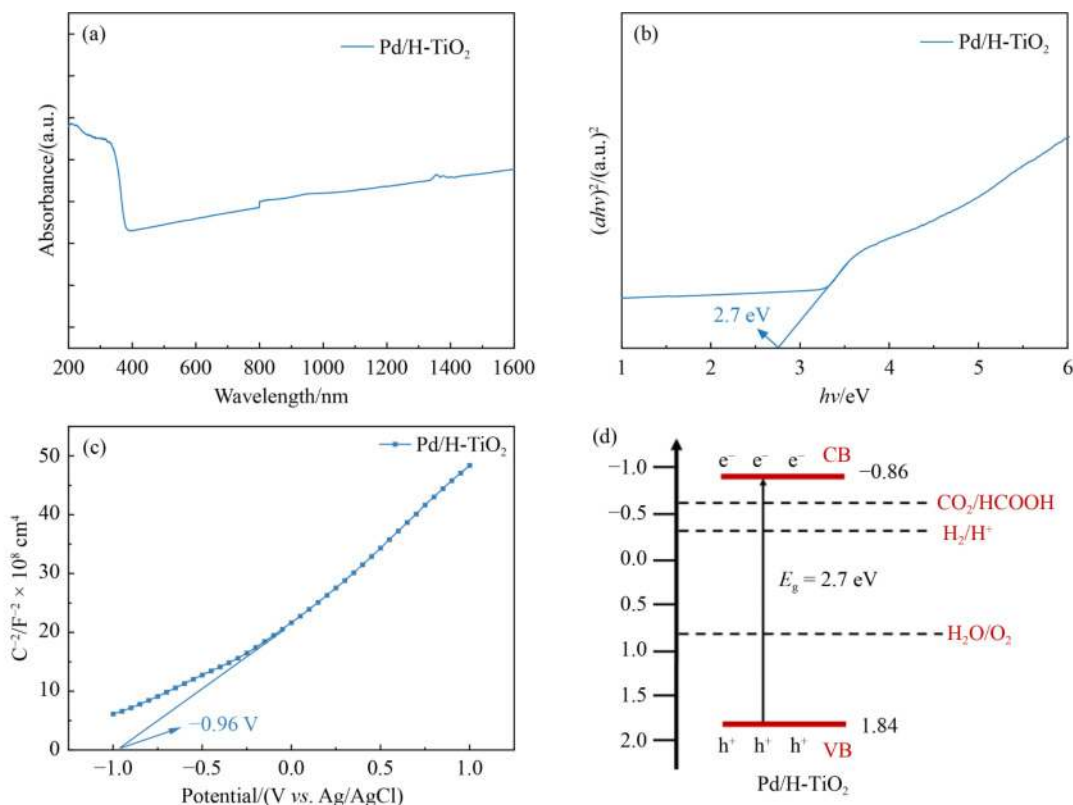


Fig. 4 (a) UV-vis spectra of various photocatalysts, (b) Tauc plots, (c) Mott-Schottky curves at 2000 Hz, (d) band structures.

adsorption energies of the models considered are listed in Table 1. The E_{ads} of CO₂ on the TiO₂ surface without O_v was approximately −0.104 eV, while the E_{ads} became more negative with the appearance of O_v on TiO₂, indicating that O_v boosted the interaction between CO₂ and the catalyst. In particular, a series of models with O_{v1} were found to have relatively stable CO₂ adsorption conformations, which was attributed to differences in O_v formation energy resulting from their different coordination environments. Although the E_{ads} of both O_{v1} and O_{v2} were negative, the experimental process was not easily achieved. Furthermore, the E_{ads} of H-TiO₂ with Pd was more negative than TiO₂ and H-TiO₂, with the exception of both O_{v1} and O_{v2} in a model. This phenomenon was attributed to the significant deformation of the Pd cluster structure caused by two O_v forms (Fig. S5(a), cf. ESM), which exhibited a higher adsorption energy value. These results demonstrated both that O_v significantly enhanced CO₂ adsorption and that Pd loading promoted adsorption and activation, agreeing with the characterization results of CO₂-TPD (Table S1).

The charge transfer during CO₂ adsorption activation process was further analyzed using charge differential density and Bader charge, as shown in Figs. 4(a–c). It is important to note that the direction of electron transfers between Pd, TiO₂, and CO₂ was not clear. Therefore, a

detailed analysis of charge transfer between adjacent species was conducted by considering differential charge density. First, the charge transfers between CO₂ and the catalyst were considered, as shown in Figs. S6(a–d) (cf. ESM), showing that CO₂ gains electrons (approximately 0.4 e), consistent with previous reports indicating that it was more difficult to lose electrons from CO₂ while electron accumulation was more easily activated. Second, CO₂ and Pd were considered in the investigation of the charge transfer between Pd and TiO₂ (Figs. S7(a–d)). The electron cloud intensity between Pd and TiO₂ was greater than that between CO₂ and Pd at an isosurface of 0.002. This indicates that the electron transfer between Pd and TiO₂ plays a crucial role in the adsorption and activation of CO₂, which had also been detected by XPS characterization. In addition, the charge of a model with both O_{v1} and O_{v2} presented transfers from TiO₂ to Pd, fitting with XPS experimental results indicating that TiO₂ has abundant oxygen vacancies after H₂ treatment.

As shown in Figs. 5(a–c) (considering O_{v12} with both O_{v1} and O_{v2} as an example), electrons were transferred from perfect TiO₂ to CO₂ (0.120 e). When O_v was present, the amount of CO₂ transferred charge also increased to −0.178 e. After Pd loading and hydrogenation treatment, the amount of CO₂ transferred charge reached −0.405 e. Pd was shown to play a very important

Table 1 Adsorption geometry and energy of CO₂ adsorbed on TiO₂ or Pd/H-TiO₂ surface

Model	$d(\text{O}_{\text{CO}_2}-\text{Ti}_{\text{TiO}_2})/\text{\AA}$	$d(\text{C}_{\text{CO}_2}-\text{O}_{\text{CO}_2})/\text{\AA}$	$d(\text{C}_{\text{CO}_2}-\text{O}_{\text{TiO}_2})/\text{\AA}$	$\angle\text{CO}_2(\text{O}-\text{C}-\text{O})/^\circ$	$d(\text{Pd}-\text{C}_{\text{CO}_2})/\text{\AA}$	E_{ads}/eV
CO ₂ -TiO ₂	2.21/2.21	1.26/1.26	1.42	135.8	–	−0.104
CO ₂ -H-TiO ₂ -O _{v1}	2.23/2.20	1.25/1.26	1.41	134.8	–	−0.357
CO ₂ -H-TiO ₂ -O _{v2}	2.20/2.30	1.26/1.24	1.45	136.9	–	−0.312
CO ₂ -H-TiO ₂ -O _{v12} ^{a)}	2.21/2.20	1.25/1.26	1.44	136.7	–	−0.379
CO ₂ -Pd/TiO ₂	–	1.23/1.22	–	145.4	2.05	−0.412
CO ₂ -Pd/H-TiO ₂ -O _{v1}	–	1.28/1.22	–	134.3	2.04	−0.885
CO ₂ -Pd/H-TiO ₂ -O _{v2}	–	1.23/1.23	–	145.4	2.01	−0.600
CO ₂ -Pd/H-TiO ₂ -O _{v12}	–	1.24/1.23	–	143.2	2.01	−0.114

a) O_{v12} represents a model containing both O_{v1} and O_{v2}.

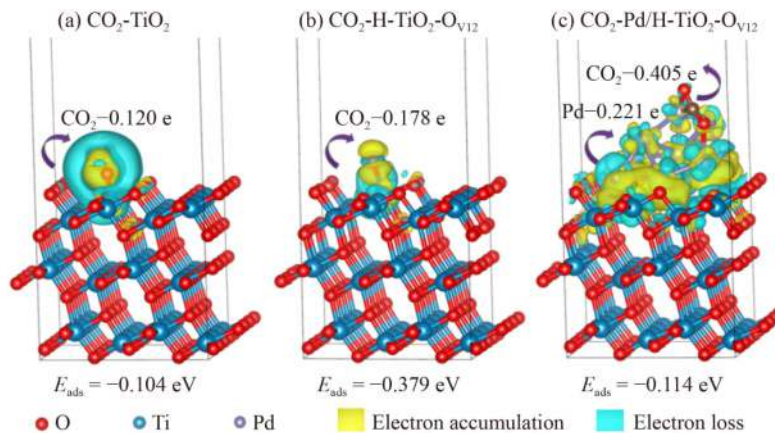


Fig. 5 (a–c) Adsorption configurations of CO₂ with and without O_v and Pd and the charge density difference upon CO₂ adsorption. Yellow and cyan represent electron accumulation and loss, respectively, isosurfaces = 0.0002 e·Bohr^{−3}.

role in this process, consistent with the above-mentioned analysis. This also indicated that O_v and Pd were crucial for CO_2 adsorption and activation, leading to excellent HCOOH yields.

3.2.3 Mechanism of the enhanced CO_2 reduction of Pd/H- TiO_2

Generally, it is considered that charge transfer efficiency is an essential influence on the photocatalytic reduction of CO_2 [12]. Photocurrent densities were measured to evaluate photoelectronic properties according to the photocurrent response and PL. The photocurrent response intensity directly reflects photogenerated charge separation efficiency [39]. Figure 6(a) shows the transient photocurrent responses of TiO_2 , H- TiO_2 , and Pd/H- TiO_2 under UV light irradiation. H- TiO_2 has a higher photocurrent density than TiO_2 , indicating that O_v performs the function of separating electron holes. In addition, the Pd/H- TiO_2 catalyst possesses the highest photocurrent density, indicating the higher separation and transfer efficiency of electron-hole pairs leading to superior HCOOH production rates. It has been reported that the rate of charge recombination ($\sim 10^{-9}$ s) is much faster than that of the surface redox process (10^{-8} to 10^{-1} s) [33], such that only electrons migrating to the catalyst surface can participate in the subsequent conversion reactions to product HCOOH. As the number of electrons on the catalyst surface increases, the CO_2 conversion rate increases, which in turn gives higher HCOOH yields. Therefore, excellent electron-hole separation ability is an important prerequisite for high photocatalytic reduction of CO_2 activity. The high photocurrent response of Pd/H- TiO_2 can be ascribed to the role of Pd as the bridge to accelerate photogenerated electron-hole transfers. The electron-hole complexation efficiency was further investigated using PL spectroscopy. Typically, the weaker PL peak represents the lower efficiency of electron-hole complexation under illumination [40,41]. As shown in Fig. 6(b), the increasing order of PL intensities under UV light is Pd/H- TiO_2 < H- TiO_2 < TiO_2 , which indicates the

presence of O_v and Pd inhibiting the direct compounding of photogenerated electron holes, consistent with transient photocurrent responses. Subsequently, the electrons transferred to the Pd can be used for CO_2 reduction [21,22].

As stated in previous reports, the role of Pd in photocatalytic reaction primarily includes: (i) the separation of electron holes; (ii) enhancing light absorption, thereby generating heat energy due to the localized surface plasmonic resonance (LSPR) effect [42,43]. The role of Pd as an electron trap for electron-hole separation has been demonstrated by PL and photoelectrochemical examinations. To investigate the LSPR effect in Pd/H- TiO_2 under UV light, infrared thermal imaging was used to collect infrared images of sample surfaces in the noncontact mode. As shown in Fig. S8 (cf. ESM), following 30 min of irradiation, the center temperature of the Pd/H- TiO_2 suspension was 26.5 °C, demonstrating that Pd played the role of separating electron-hole under UV light.

3.3 Mechanism of photocatalytic reduction of CO_2 in aqueous systems

We performed diffuse-reflectance infrared Fourier transform spectroscopy (DRIFTS) to further analyze the processes of CO_2 adsorption and activation. Intermediate products of CO_2 conversion in the presence of the Pd/H- TiO_2 catalyst were detected at different times, as shown in Fig. 7. The peaks centered at 1035 and 934 cm^{-1} were ascribed to the adsorbed $*HCO_3$ and $*CO_3$, respectively; implying that some CO_2 was converted to HCO_3^- or CO_3^{2-} [44,45]. The characteristic absorption peaks located at 2200–2400 cm^{-1} were ascribed to CO_2 adsorbed onto the Pd/H- TiO_2 catalyst, implying strong CO_2 adsorption on the Pd/H- TiO_2 surface. Peaks centered at 1679 and 1392 cm^{-1} were assigned to the $\nu_{C=O}$ stretching vibration of adsorbed HCOOH and the symmetric stretching vibration of COO^- in $*COOH$ [46].

When inorganic bases and water were used as the reaction system, CO_2 was able to lead to the derivation of multiple intermediates, including carbonate and

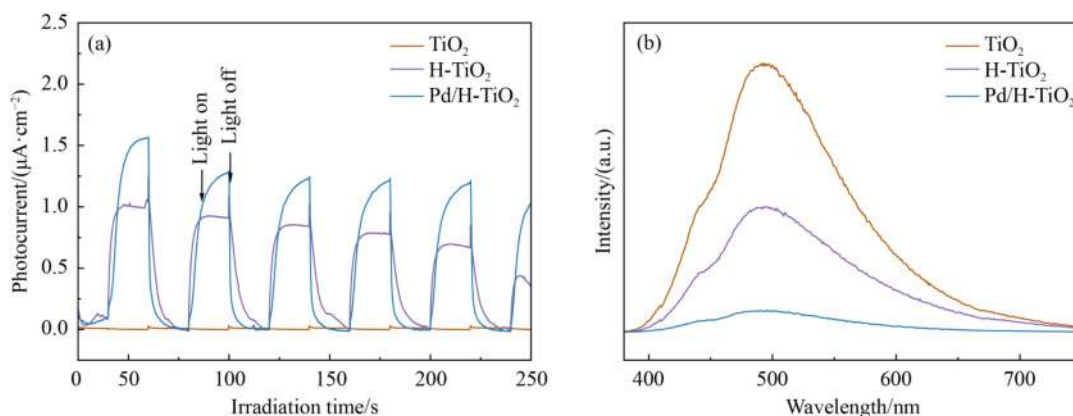
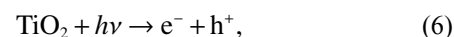


Fig. 6 (a) Transient photocurrent response, (b) PL spectra of TiO_2 , H- TiO_2 , and Pd/H- TiO_2 .

bicarbonate. To verify whether HCO₃⁻ and CO₃²⁻ participated in the reaction, bicarbonate and carbonate were used in the photocatalytic reaction (Fig. S9, cf. ESM). Our results showed that a small amount of HCOOH was produced. From the above results, it can be concluded that there are three intermediate species in photocatalytic reduction in the presence of a Pd/H-TiO₂ catalyst. The process may include the adsorption of CO₂ onto a TiO₂ surface, which quickly reaches equilibrium, causing some of the CO₂ to be converted to CO₃²⁻ and HCO₃⁻, which then participate in the reaction.

Combining the characterization, DFT calculation, and experimental analyses outlined above, a reaction mechanism was proposed as illustrated in Fig. 8. The reaction stage of photocatalytic reduction of CO₂ to HCOOH contained the CO₂ adsorption activation stage on the surface and the reduction reaction stage. First, CO₂ and H₂ molecules were co-adsorbed onto the surface of the catalyst. TiO₂ was excited under UV irradiation to produce electron-hole pairs to participate in the reaction (Eq. (4)). Photogenerated electron-hole pairs were then transferred to the surface of the catalyst in order to activate CO₂ and H₂ molecules to form active species of

*CO₂ and H [47]. Subsequently, *COOH intermediates formed by the transfer of H atoms in H₂ to O atoms of CO₂ led to their conversion from intermediates to the target product of HCOOH (Eq. (5)). H₂O molecules were oxidized by h⁺ to form OH groups (Eq. (6)).



4 Conclusions

The Pd/H-TiO₂ catalyst was successfully prepared via a facile impregnation method combined with a hydrogen treatment process. In the presence of an appropriate base, the yield of HCOOH over a Pd/H-TiO₂ catalyst reached 14.14 mmol·g_{cat}⁻¹·h⁻¹ and the HCOOH selectivity was close to 100%. Experimental characterizations and theoretical calculations demonstrated that this superior activity was derived from the synergistic effect of Pd and O_v. In particular, O_v could increase CO₂ adsorption, while Pd loading was beneficial for electron-hole pair separation. The transfer of electrons from the support to Pd played a crucial role in promoting CO₂ activation.

Competing interests The authors declare that they have no competing interests.

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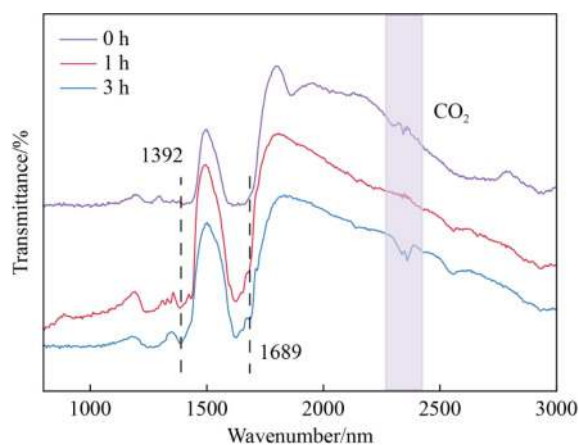


Fig. 7 DRIFTS spectra of the Pd/H-TiO₂ catalyst after the photocatalytic reaction.

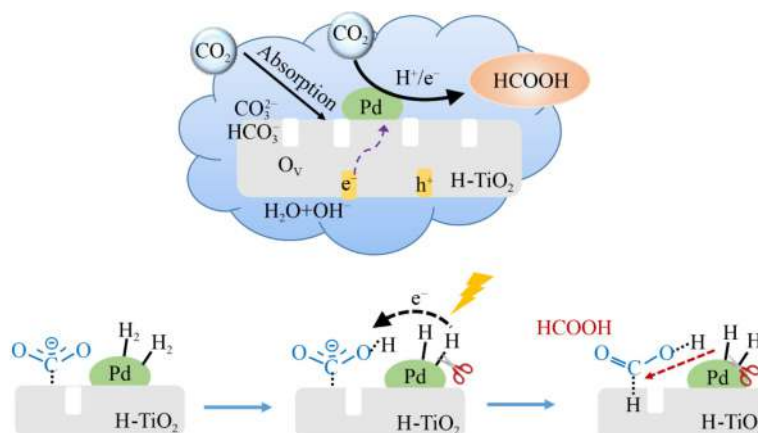


Fig. 8 Mechanism of the photocatalysis of CO₂ to produce HCOOH.

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